

MP5 — FINAL PORTFOLIO

MiniClaw Gripper

Design, Verification, and What I Learned to Distrust

A force-efficient four-bar gripper, **verified three independent ways.**

GOAL & DIRECTION

The brief, and the choice that shaped everything

Scale the BigClaw down: 0–86 mm jaw → ~40 mm

MiniClaw target.

Fit one side inside the 46 × 55 mm half-envelope.

Other side mirrors; total jaw = 2 × single-side displacement.

THE DESIGN DRIVER I CHOSE

Force efficiency

Keep the transmission angle near 90° across the whole stroke — **not just chase maximum opening.**

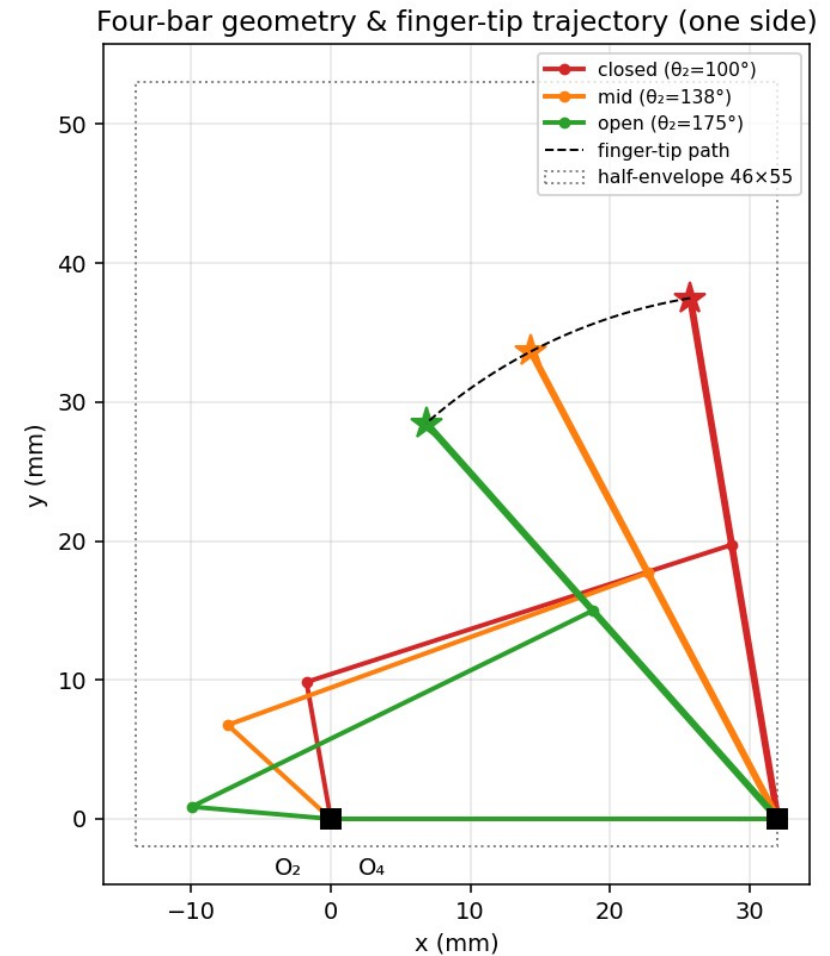
This one decision is why the linkage came out the way it did.

THE DESIGN

Grashof crank-rocker, one side

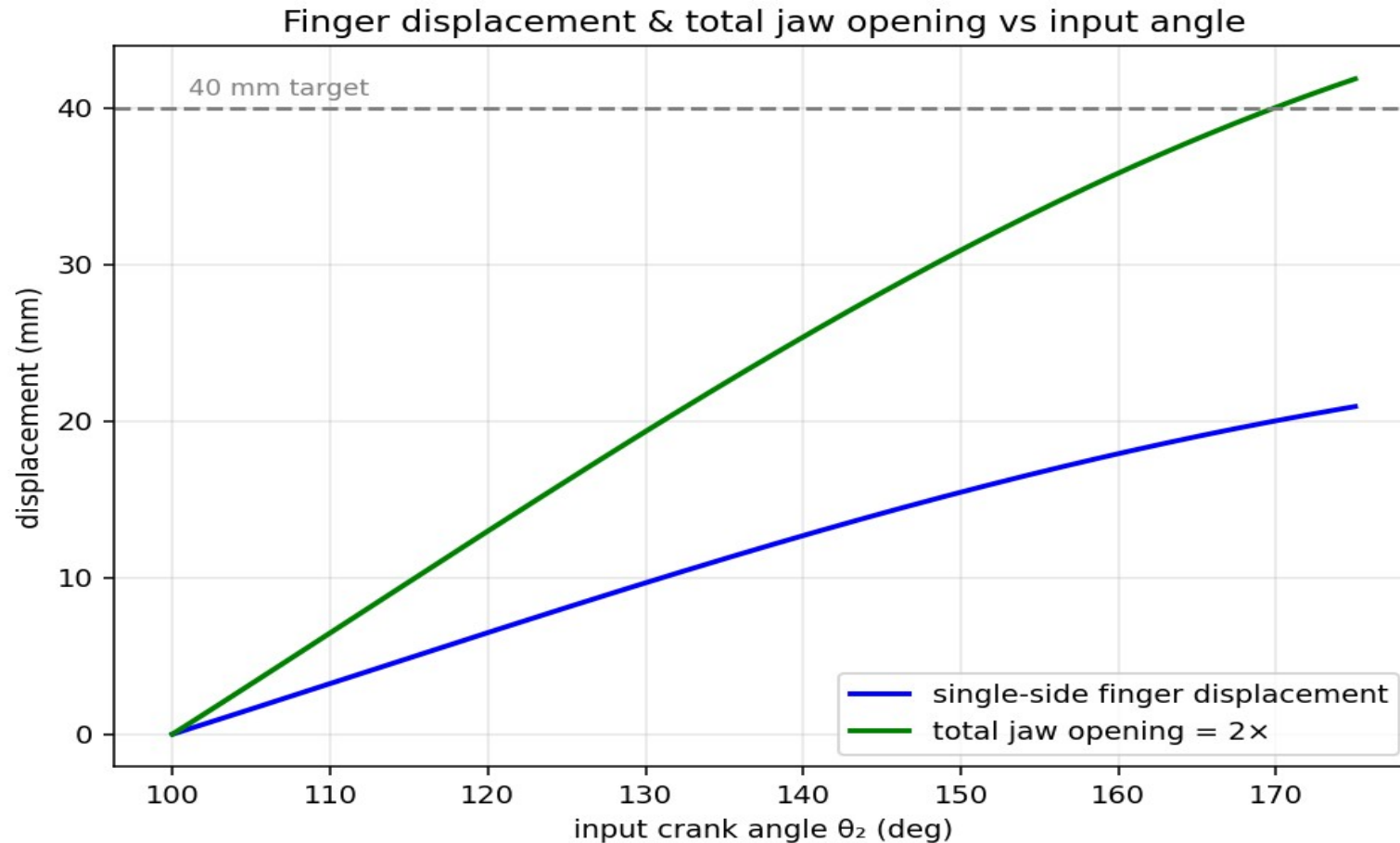
Ground r_1	32 mm
Input crank r_2	10 mm
Coupler r_3	32 mm
Output rocker r_4	20 mm
Finger from O_4	38 mm
Input sweep	$100^\circ \rightarrow 175^\circ$ (75°)

Grashof crank-rocker — input crank fully rotatable, drivable by a continuous gear train.



ANALYSIS 1 — POSITION

0 → 41.9 mm jaw, monotonic



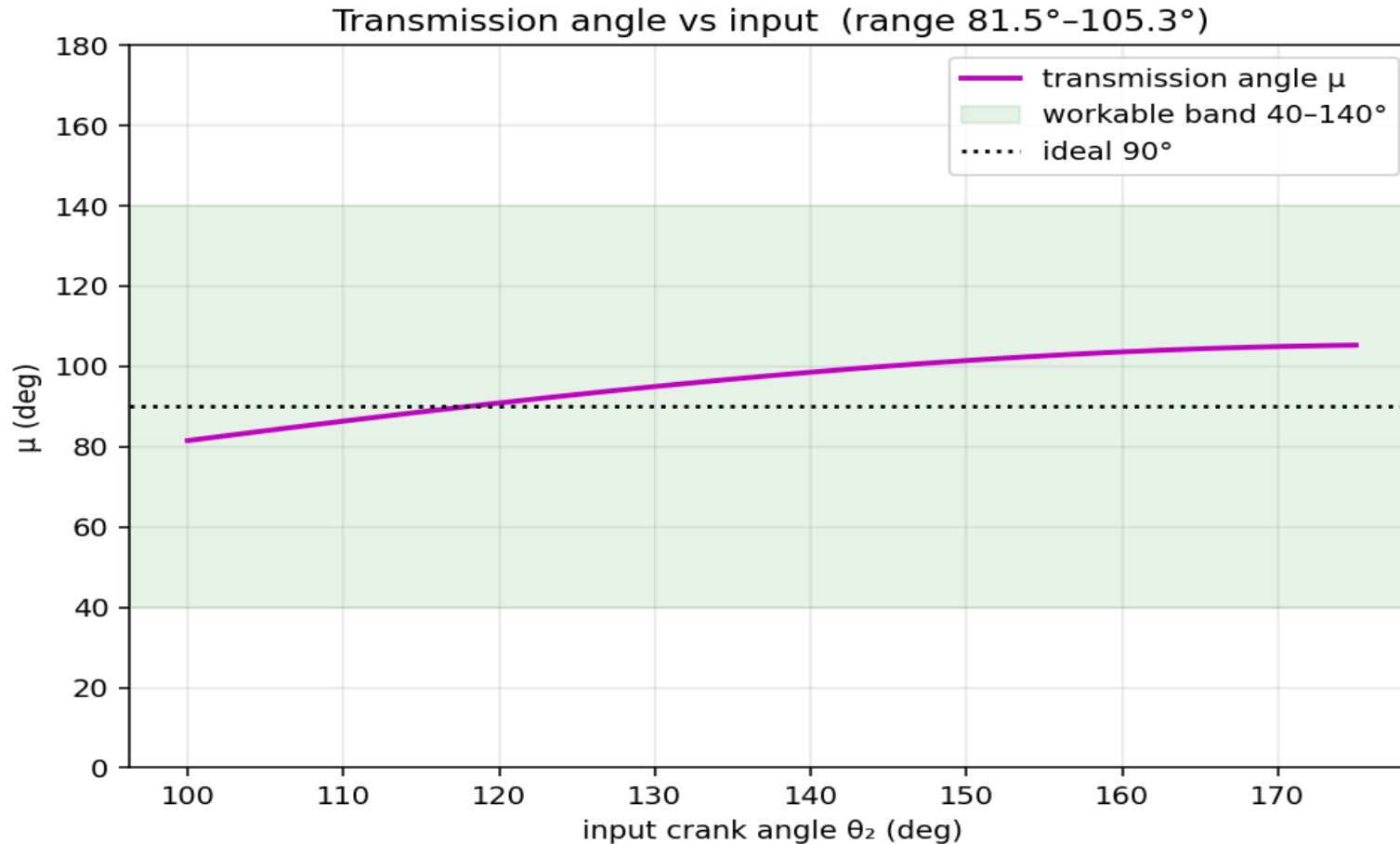
41.9 mm

total jaw opening

Single-side: **20.9 mm**

*Monotonic — no motion reversal
across the sweep.*

Why I picked this geometry

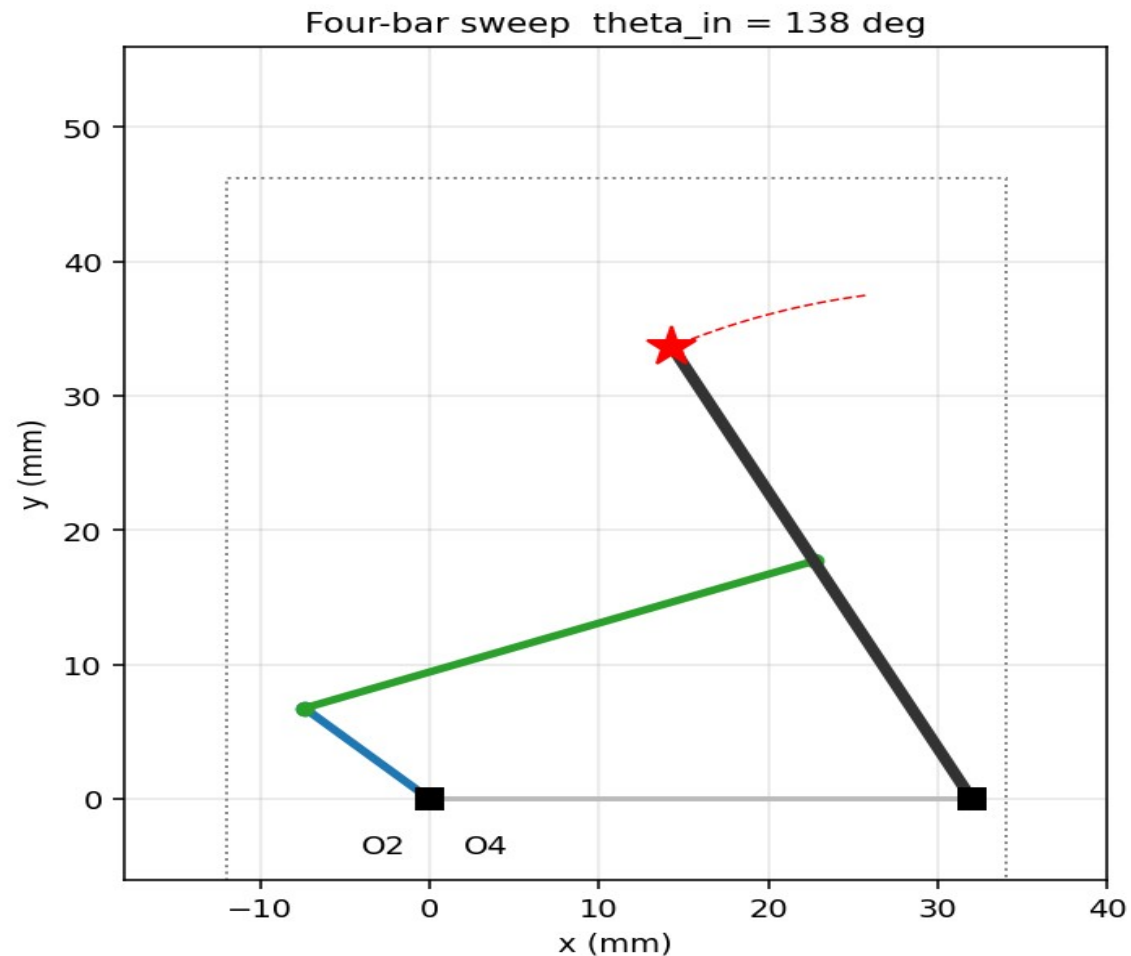


81.5°–105.3°

μ across full sweep

Straddles the ideal 90°.
Never within 38° of a singularity —
strong, uniform output force
everywhere.

Nothing binds, everything fits



5.18 mm

minimum link-to-link clearance — no crossings

2.02 mm

minimum wall clearance — fits the half-envelope (tight on width)

Interactive demo plays the full open → close cycle.

Triangulation: three paths, one answer

Hand calc

Closed-form circle
intersection, 3 poses

Simulation

240-step sweep,
motion artifact

AI analysis

Parameter sweep +
review loops

All three agree to < 0.1 mm on tip position and to the same μ band. The < 0.1 mm gap is rounding, not disagreement.

If they ever disagreed, I trust the hand calc — the closed form is exact for matched link lengths, so the function would be the thing to debug.

When the AI was confidently wrong

The first AI parameter sweep handed me a linkage that **hit the 40 mm jaw target** — but let the transmission angle drop to **~58° near full open**.

It passed a naive “is it in the 40–140° band?” check, so it looked acceptable. I caught that it had optimized stroke in isolation, forced a re-run with a μ penalty, and converged on 32/10/32/20 — μ 81.5–105.3°.

THE LESSON

The AI satisfied the letter of the spec while missing its intent.

Knowing the real acceptance criteria is the engineer's job.

Drive train: 12:1, synchronized

Architecture	Compound spur + 1:1 sync pair
Reduction N	12 : 1 (2.5 thumb turns → 75° sweep)
Stage 1	12T → 48T (4:1)
Stage 2	16T → 48T (3:1)
Sync pair	40T ↔ 40T (counter-rotating)

COUPLING CHECK

N doesn't change the linkage
— it only maps thumb turns to
the same 75° sweep.

✓ μ stays 81.5–105.3° —
reduction keeps the linkage
in band.

What I would not send to the print queue tomorrow

m = 1.0 mm gear teeth

Root fillet ~0.38 mm is below the comfortable FDM feature. Print one test gear first; m = 1.25 fallback ready.

27 parts vs. < 15 target

Drive train drives the count. Merges named for MP5: combine stage gears, single-shell housing.

2 mm envelope margin

Tight on width — must be re-confirmed in 3D once wall thickness and pin heads are added.

Final trust ledger

WHAT I TRUST

- μ 81.5–105.3° across the full sweep (hand calc + sim)
- 41.9 mm jaw, monotonic, agree to < 0.1 mm
- No link crossings — 5.18 mm min clearance
- Fits 46×55 half-envelope; Grashof, drivable
- Drive train N=12:1, 1:1 sync, coupling check passes

WHAT STILL NEEDS CHECKING

- Pin clearance & friction (ideal-joint model)
- Input range depends on N — re-check if N changes
- Symmetry holds only if sync built & verified
- 2 mm width margin in 3D stack-up
- Gear tooth strength / backlash at m=1.0
- Part count 27 vs. <15 target

It moves — and the AI runs live

1. **Interactive linkage simulator** (runs in the browser) — drag the slider; both jaws mirror and the gripper opens and closes.

Live readouts: input angle, transmission angle (in-band badge), and jaw opening 0 → 41.9 mm.

2. **Live MCP / tool call** on a real design question:

“ $m=1.0$, $z=12$ pinion — root fillet radius, printable on a 0.4 mm nozzle?”

It moves and computes the verified physics live — no CAD dependency. Backup: screen recording + recorded tool-call run.

THE DURABLE LESSON

**The tools will be replaced.
The skill won't.**

Knowing what a “good” answer actually has to satisfy — and auditing the AI against the constraint it quietly skipped — **is what transfers to any mechanism and any tool I'm handed next.**

Thank you · Questions?